

Modeling Attempt-to-Escape Frequency: Tunneling Emission of Trapped Electrons in Tunneling Oxides of 3-D NAND Flash Memory

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Abstract—We propose a novel physical model for the attempt-to-escape frequency of trap-to-band electron emission, which is broadly applicable to various trap-to-band scenarios. The model is verified under detrapping mechanisms in bandgap-engineered tunneling oxide (BETOX), enabling accurate prediction of electron emission dynamics within extremely short timeframes. Extensive comparisons between the proposed model and calibrated TCAD simulations demonstrate excellent agreement, validating its accuracy and reliability. Additionally, based on calibrated physical parameters, the model is adaptable to engineering variations such as trap profiles, including intricate combinations of Gaussian trap distributions, making it highly versatile for future device optimization and analysis.

Index Terms—Attempt-to-escape frequency, bandgap-engineered 3-D charge trap NAND flash memory, TCAD simulation, trap-to-band tunneling (TBT) emission.

I. INTRODUCTION

THE growing demand for advanced memory solutions, driven by AI and high-performance computing (HPC), has positioned 3-D NAND flash as a key innovation in the semiconductor industry [1]. By stacking memory cells vertically, 3-D NAND enables higher capacity and lower bit cost compared to 2-D NAND [2]. Recent advancements in bandgap engineering of tunneling oxides through nitride incorporation have improved performance by widening the memory window [3]. However, this also leads to challenges such as trap site formation within the tunneling oxide, which can result in retention failure caused by electron detrapping [4].

The development of physical compact models for retention failure in 3-D NAND flash memory has been extensively explored through various approaches [4], [5]. These models aim at accurate simulations of retention failure in 3-D

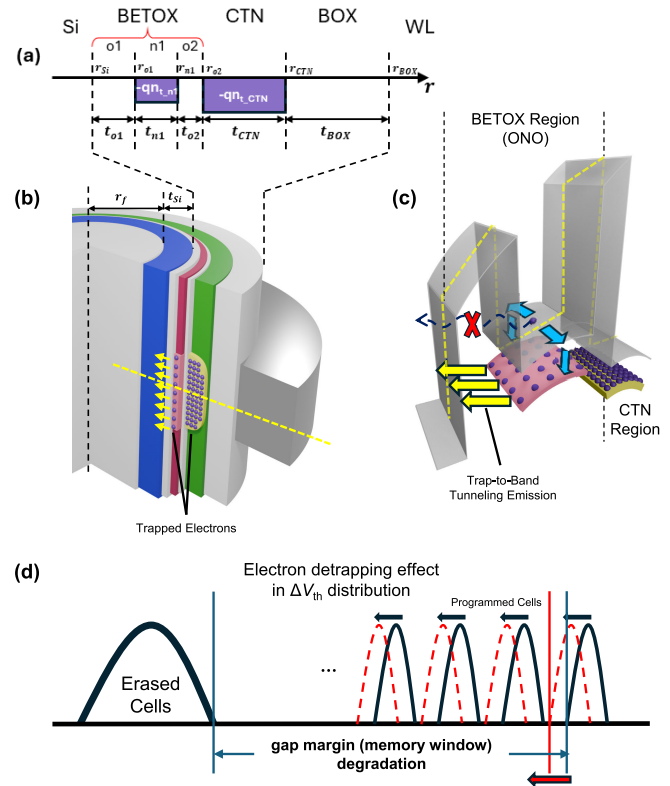


Fig. 1. Configuration of the (a) charge distribution by radial coordinate, (b) 3-D structure, (c) 3-D band diagram of the BETOX 3-D NAND flash indicating dominant emission mechanism as TBT, and (d) effect on reliability by electron detrapping in 3-D NAND flash memory cell ΔV_{th} distribution.

NAND flash memories, but often rely on simple shockley-read-hall (SRH) theory [6], [7], while tunneling emission needs investigation of carrier traps and reservoirs at different regions.

In this study, we propose a physically accurate model for the attempt-to-escape frequency for trap-to-band tunneling (TBT) emission (ν_{TBT}) of trapped electrons and apply it to electron detrapping within bandgap-engineered tunneling oxide (BETOX) [4], [8]. Furthermore, this study explores the impact of various trap profile distributions, to better capture their inherent characteristics.

Received 27 May 2025; revised 23 June 2025 and 8 July 2025; accepted 11 July 2025. Date of current version 25 August 2025. The review of this article was arranged by Editor S.-M. Hong. (Corresponding authors: Myung Jin; Hyungcheol Shin.)

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Digital Object Identifier 10.1109/TED.2025.3589340

II. SETUP AND DEVICE PARAMETERS

A. Calibration of the Simulation and Model

Fig. 1 summarizes the structural, band diagram, and the effect of electron detrapping in BETOX 3-D NAND flash memory cells. For the simulation constructed throughout this study, Synopsys Sentaurus TCAD was used, which can conduct an accurate 3-D semiconductor simulation [9]. After constructing the 3-D cylindrical structure and adopting the physical parameters as Table I, the program operation by a single pulse of duration 1 μ s and rising and falling times in 0.1 μ s, then the retention operation was continued by setting the voltage of all the current contacts to 0 V. The programming of the cells was done after the erase of the whole NAND flash string. Over the whole simulation, nonlocal tunneling was implied for the physics of the tunneling mechanism within the device structure [9].

Fig. 1(b) shows the programmed cell with trapped electrons within the charge trap nitride (CTN) layer and BETOX nitride ($n1$) layer. All physical parameters, structural dimensions, and V_{pgm} values organized in Table I were calibrated through TCAD simulations based on previous studies [10], [11], [12]. The program operation was employed to all word lines uniformly programmed to achieve identical threshold voltage shifts (ΔV_{th}) as the target cell. Note that ΔV_{th} value is the difference of threshold voltage from the virgin cell ($\Delta V_{\text{th,ERS}} < 0$ V). The program voltage (V_{pgm}) was precisely calibrated using the TCAD simulation parameters listed in Table I.

After the construction of the TCAD simulation, calibration of the model to the TCAD simulation in accordance with the band diagram and electrostatic properties shown in Figs. 1(a), (c), and 2(c) analytically is needed. The following Poisson equation is solved under cylindrical structure for the model calibration [4], [13]:

$$\frac{\partial^2 V(r)}{\partial r^2} + \frac{1}{r} \frac{\partial V(r)}{\partial r} = \begin{cases} -\frac{qn_{t,n1}}{\epsilon_{n1}} & \text{if } r \in n1 \\ -\frac{qn_{t,CTN}}{\epsilon_{CTN}} & \text{if } r \in \text{CTN} \\ 0 & \text{if } r \in \text{else} \end{cases} \quad (1)$$

where $n_{t,n1}$ and $n_{t,CTN}$ represent the trapped electron density of $n1$ and CTN regions, respectively, which is assumed to have a uniform concentration distribution, as shown in Fig. 1(a). Here, q denotes the absolute value of a single electron charge. The detailed solution of the Poisson equation in the BETOX structure is provided in the Appendix.

The trapped electron within the region assumed as uniform distribution forms a difference in threshold voltage (ΔV_{th}) as in the following equation, where C_{n1} and C_{CTN} are the capacitive constants for $n1$ and CTN regions, respectively:

$$\Delta V_{\text{th}} = \frac{qn_{t,n1}}{C_{n1}} + \frac{qn_{t,CTN}}{C_{\text{CTN}}}. \quad (2)$$

Therefore, the program cell information is stored within the cell as ΔV_{th} by program operation, which injects electrons into the CTN layer. Note that $n_{t,CTN}$ plays the main role of forming the ΔV_{th} value, while $n_{t,n1}$ is formed due to unintentional electron trapping during the program operation.

TABLE I
PARAMETER VALUES FOR THE TCAD SIMULATION

Symbol	Physical meaning	Value
r_f	Oxide filler radius	20 nm [10]
t_{Si}	Si-channel thickness	8 nm [10]
$t_{o1}/t_{n1}/t_{o2}$	BETOX oxide thickness O/N/O	2/2/0.9 nm [10]
t_{CTN}	Charge trap nitride thickness	6 nm [10]
t_{BOX}	Blocking oxide thickness	8 nm [10]
ϵ_{ox}	Oxide permittivity	$3.9 \epsilon_0$ [11]
ϵ_{n1}	BETOX nitride permittivity	$6.0 \epsilon_0$ [10]
ϵ_{CTN}	Charge trap nitride permittivity	$7.5 \epsilon_0$ [11]
$\Delta\Phi_{Si-ox}$	Si-ox layer energy barrier height	3.1 eV [10]
$\Delta\Phi_{n1-ox}$	$n1$ -ox layer energy barrier height	0.5 eV [10]
$\Delta\Phi_{CTN-ox}$	CTN-ox layer energy barrier height	1 eV [10]
$\Delta\Phi_{WL-ox}$	WL-ox layer energy barrier height	3.65 eV [10]
m_{ox}^*	The effective mass of oxide	$0.5 m_0$ [11]
m_n^*	The effective mass of nitride	$0.4 m_0$ [11]
σ_n	BETOX electron capture cross section	$1 \times 10^{-16} \text{ cm}^2$ [12]
V_{pass}	Pass voltage for untarget cell	7.0 V [10]
$V_{\text{pgm},0}$	Program voltage for $\Delta V_{\text{th}} = 0$ V	16.7 V
$V_{\text{pgm},1}$	Program voltage for $\Delta V_{\text{th}} = 1$ V	18.1 V
$V_{\text{pgm},2}$	Program voltage for $\Delta V_{\text{th}} = 2$ V	19.4 V

Fig. 2(a) depicts the uniformly patterned cells. Since the trapped electrons within the CTN layer are effectively equivalent to a negative gate voltage given to the NAND string, the Si-channel becomes depleted. This makes the string an insulated area, forming an extra negative channel potential (V_{ch}) to the area of the programmed cell [14]. Fig. 2(b) shows that the value of $V_{\text{ch}} \approx -\Delta V_{\text{th}}$ as observed within the TCAD simulation. Fig. 2(c) depicts the Poisson equation over cross-sectional view.

Since it has been reported that retention failure due to trapped electron emissions within tunneling oxide could affect the reliability of the 3-D NAND flash cell in both short-term and long-term timescales [4], [8], the time transient of $n_{t,n1}$ was analyzed precisely throughout this study.

B. Assumption of Dominant Emission Mechanism

As illustrated in Fig. 1(c), we assume TBT as dominant mechanism during electron detrapping from BETOX since electrons in the conduction band are more probable to be recaptured to available trap sites because electrons cannot excessively accumulate to enable fowler-nordheim (FN) tunneling or direct tunneling [15], as described in Fig. 2(e).

Conduction band diagram of BETOX can be induced, as shown in Fig. 2(c) using the Poisson equation solution in the Appendix. Fig. 2(d) and (e) shows the difference of electronic statistics between program and retention operation. Program operation as Fig. 2(d) generates the FN tunneling current (J_{FN}) by a high field with intensively accumulated conductive electrons in the conduction band region.

However, during a retention operation, as illustrated in Fig. 2(e), the Fermi level (E_F) is positioned near the average trap depth due to the relaxation nature of a retention operation. This energy band configuration prevents excessive electron accumulation in the conduction band of BETOX $n1$ layer. Consequently, trapped electrons are more likely to be emitted directly to the conduction band of the Si channel rather than being transported through direct tunneling to the channel region. Therefore, the process of trapped electron detrapping from the BETOX region is predominantly governed by TBT emission.

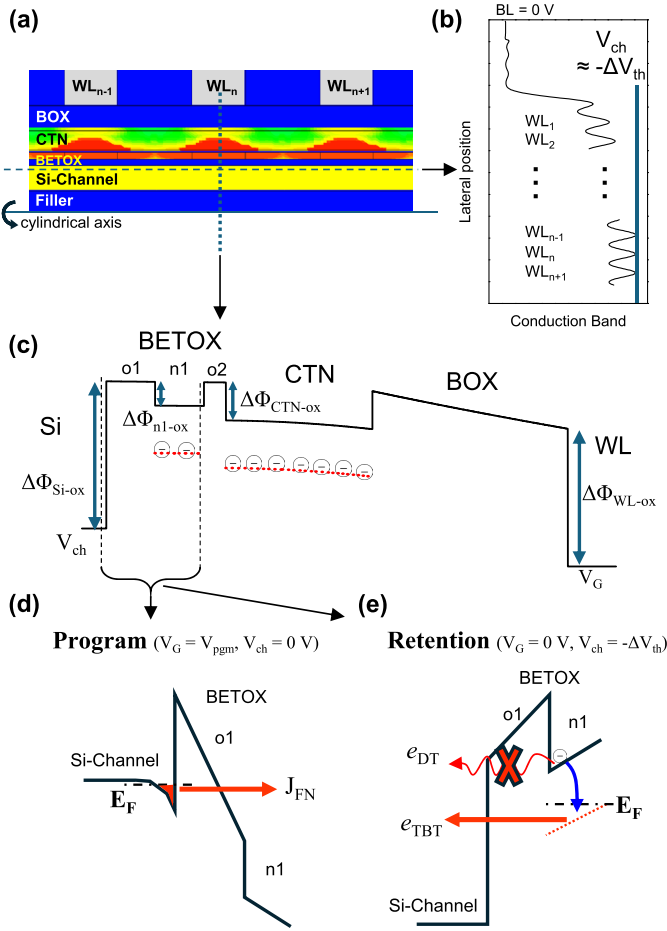


Fig. 2. (a) TCAD simulation setting of cylindrical 3-D NAND flash memory string. The conduction band profile of (b) lateral slice and (c) cross-sectional view. The main tunneling mechanism of electrons in (d) program operation and (e) retention operation.

The temporal evolution of trapped electron density can be modeled by considering various emission mechanisms. However, based on both theoretical analysis and experimental observations [4], [8], the TBT mechanism dominates the emission process during retention failure caused by electron detrapping mechanisms. Therefore, the emission rate equation can be simplified as

$$\frac{dn_{t,n1}}{dt} = \sum_m^{\text{mechanisms}} -e_m n_{t,n1} \approx -e_{\text{TBT}} n_{t,n1} \quad (3)$$

where e_m denotes the emission rate of various mechanisms, and e_{TBT} is the TBT emission rate. While multiple emission pathways potentially contribute to retention loss, (3) reflects our finding that a single dominant mechanism—TBT—adequately describes the observed electron emission behavior in the BETOX structure.

III. MODEL DESCRIPTION

A. Band-Trap Electron Transition Model

The tunneling emission mechanism is rigorously modeled through quantum state transitions between the Si-channel conduction band and trap sites, as schematically depicted in Fig. 3(a). The Si-channel conduction band functions as an electron reservoir that dynamically interacts with

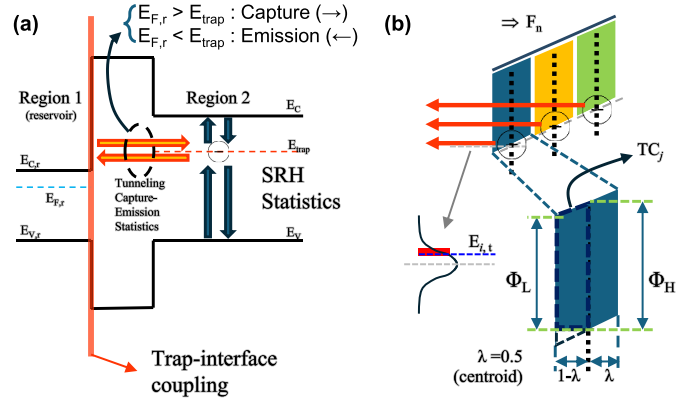


Fig. 3. (a) Electron transition mechanism between Si-channel conduction band and trap sites in BETOX structure. (b) Discretized representation of continuous trap energy profiles with multibarrier tunneling paths.

nitride-induced trap states within the BETOX layer. The fundamental transition directionality arises from Fermi level (E_F) alignment disparities—electrons migrate from regions of higher Fermi potential (trap region) to lower Fermi potential (Si-channel region) under retention bias conditions.

Fig. 3(b) explicitly illustrates the multibarrier energy landscape governing electron emission from continuously distributed trap profiles. To enable computationally tractable modeling, spatial uniformity of trap distribution within the BETOX layer is assumed, as shown in Fig. 1(a), allowing the discretization of the continuous trap profile into finite energy and spatial resolutions during numerical simulations.

The emission kinetics for individual trap levels are governed by first-order ordinary differential equations (ODEs), with the general solution of (3) expressed as

$$n_t = \sum_{i=1}^{\text{trap spatial}} \sum_{j=1}^{\text{trap spatial}} n_{t,i,j}(t) = \sum_{i=1}^{\text{trap spatial}} \sum_{j=1}^{\text{trap spatial}} n_{t0,i,j} \exp(-e_{\text{TBT},j} t) \quad (4)$$

where index i enumerates the discretized trap energy levels and j denotes the spatial partitioning within the BETOX layer. Enhanced simulation fidelity is achieved through refined discretization of both energy (i) and spatial (j) domains though this improvement necessitates proportional increases in computational resource allocation.

This formulation provides a unified framework for predicting time-dependent charge loss in BETOX-based memory devices, explicitly coupling quantum mechanical tunneling probabilities with macroscopic device electrostatics. The model's discretization approach enables systematic investigation of trap profile engineering effects on retention characteristics while maintaining compatibility with industry-standard TCAD simulation platforms.

B. Attempt-to-Escape Frequency (ν_{TBT}) Modeling

Evaluating the TBT emission rate (e_{TBT}) as depicted in Fig. 3, which is defined as

$$e_{\text{TBT}} = \nu_{\text{TBT}} \times |\text{TC}| \quad (5)$$

represents the emission rate as multiplication of the attempts to escape (ν_{TBT}) with the transmission coefficient ($|\text{TC}|$) of multiple trapezoidal barriers as Fig. 4(b).

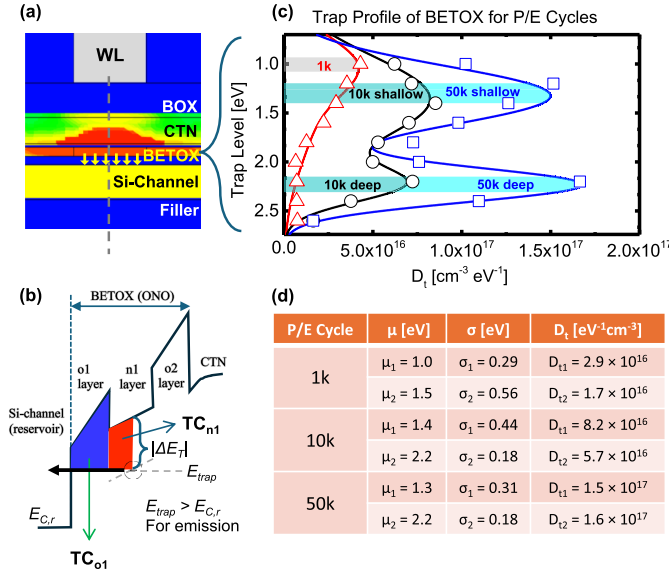


Fig. 4. (a) TCAD representation of electron detrapping within the device. (b) Multibarrier electron detrapping by the energy band cross-sectional view. (c) Gaussian fitting of the measured BETOX trap profile [10]. (d) Specification.

The $|TC|$ value is calculated using the Wentzel–Kramers–Brillouin (WKB) approximation [16], which accounts for the cumulative effect of multiple trapezoidal potential barriers between the trap and the conduction band. As depicted in Fig. 3(b), the tunneling probability for each individual barrier is expressed in exponential form, and the overall transmission coefficient is obtained by multiplying the transmission probabilities of all the barriers

$$|TC| = \prod_{o1, n1} \exp\left(-\frac{4\sqrt{2m_{ox,n}^*}}{3\hbar q} \cdot \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F_{eq}}\right). \quad (6)$$

Here, Φ_H and Φ_L represent the higher and lower sides of the trapezoidal potential barrier, respectively, and F_{eq} denotes the equivalent electric field. F_{eq} value can be calculated with the centroid field intensity of each $o1$ and $n1$ layers provided in the Appendix. The WKB approximation is well-suited for modeling quantum tunneling in structures where the potential barrier profile varies gradually, as is the case for the multilayered insulating regions in BETOX. This approach effectively captures the cumulative tunneling probability across the multiple barriers between the trap and the conduction band.

The modeling of the attempt-to-escape frequency of TBT emission (ν_{TBT}) begins when the trap energy level is higher than the conduction band energy level of the electron reservoir ($E_{trap} > E_{C,r}$). By utilizing the tunneling capture-emission model proposed in [17], the ν_{TBT} can be derived as follows:

$$\nu_{TBT} = \int W(E) \cdot D(E) \cdot (1 - f_r(E)) dE. \quad (7)$$

The transition rate per density of states (DOS) $W(E)$ is derived using Fermi's golden rule under the influence of a spatially localized perturbation Hamiltonian $\hat{H}'_T$. Since the previous study says that amorphous compound of silicon and nitride generates hole and electron trap site by over- and under-coordinate atoms [18], the assumption of the trap volume as average intermolecular volume has been adopted. The average

amorphous Si-N compound bonding length is 1.7 Å; therefore, the assumption can lead to the trap volume of the electron trap site as $V_T = (1.7 \text{ Å})^3$.

The Hamiltonian is explicitly defined as

$$\hat{H}'_T(\mathbf{r}) = \begin{cases} -|\Delta E_T|, & \text{for } \mathbf{r} \in V_T \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where $|\Delta E_T| \equiv |E_C - E_{trap}|$ represents the energy depth of the trap relative to the conduction band edge E_C , and $\mathbf{r}$ denotes the spatial coordinate.

The matrix element $\langle f | \hat{H}'_T | i \rangle$ simplifies under the assumption of localized trap states, yielding

$$W(E) = \frac{2\pi}{\hbar} |\langle f | \hat{H}'_T | i \rangle|^2 \rho(E) = \frac{2\pi}{\hbar} |\Delta E_T|^2 V_T \delta(E - E_{trap}) \quad (9)$$

where the Dirac delta function $\delta(E - E_{trap})$ enforces energy conservation during the trap-to-band transition. This formulation directly links the microscopic trap geometry (V_T) and engineering defect profile ($|\Delta E_T|$) to macroscopic emission kinetics, providing a quantum-mechanically rigorous foundation for predicting state transition dynamics in BETOX-based devices.

The DOS for TBT in the nitride layer incorporates anisotropic effective mass components, with distinct contributions from in-plane (m_0) and tunneling-direction (m_n^*) dynamics. Spin degeneracy and quantum phase-space normalization are explicitly resolved through the derived formulation

$$D(E) = D_{||\text{tun}} D_{\perp\text{tun}}(E) = 2 \cdot \frac{1}{4\pi^2} \frac{2m_0}{\hbar^2} \frac{\sqrt{2m_n^*}}{\hbar} \sqrt{E - E_{C,r}}. \quad (10)$$

This formulation decomposes the effective mass into tunneling-direction and tunneling-surface components, reflecting the coupling between the constant trap-induced perturbation and the conduction band of the electron reservoir. Notably, the DOS integration occurs in the energy level of the Si channel (electron reservoir), as the trap in the $n1$ region acts as a constant perturbation coupled to the reservoir's conduction band.

The $(E - E_{C,r})^{1/2}$ term reflects energy-dependent state availability above the conduction band edge, while the $\hbar^{-3}$ scaling ensures proper quantum-mechanical normalization. Compared to isotropic models, this anisotropic treatment more accurately predicts emission kinetics in BETOX structures, where directional mass variations critically influence tunneling probabilities.

$f_r(E)$ is a Fermi distribution of electrons within the conduction band of reservoir. Using the parameter set as Table I, the condition $E_{trap} \gg E_{C,r}$ is satisfied, leading to $f_r(E_{trap}) \approx 0$, which implies $1 - f_r \approx 1$.

By applying state transition rate $W(E)$ and DOS value $D(E)$ to (7), ν_{TBT} is derived as follows:

$$\nu_{TBT} = \frac{2\pi}{\hbar} |\Delta E_T|^2 V_T \times \frac{\sqrt{2m_n^*} m_0}{\pi^2 \hbar^3} \sqrt{E_{trap} - E_{C,r}}. \quad (11)$$

Thus, the attempt-to-escape frequency (ν_{TBT}) has been successfully derived, conducting the subsequent emission rate (e_{TBT}) calculation combining (6) and (11) to (5).

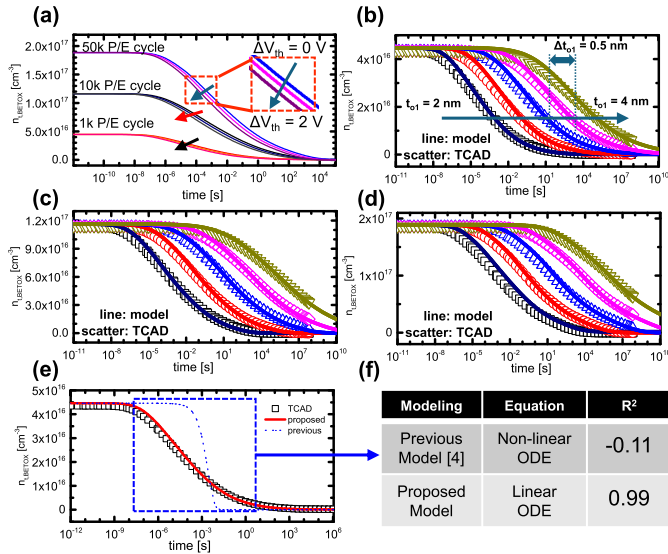


Fig. 5. (a) Comparison of model over $\Delta V_{\text{th}} = 0\text{--}2$ V with various trap profiles. (b)–(d) Compensation result of TCAD and model with trap profile of 1k, 10k, and 50k P/E cycle cells with t_{01} variation over 2–4 nm. (e) and (f) Comparison and development list of the proposed model from the previous model.

Fig. 4(c) presents the Gaussian fitting of trap energy profiles in the BETOX structure, measured experimentally using the trap spectroscopy by charge injection and sensing (TSCIS) method [10]. The continuous trap distribution derived from this fitting spans the entire bandgap energy range. To enable numerical simulation, this profile is further resolved into discrete components through double-Gaussian decomposition, as detailed in Fig. 4(d). The decomposed representation separates the trap population into two distinct Gaussian distributions, capturing both shallow and deep trap characteristics within the BETOX layer. This discretization process allows the continuous energy spectrum to be partitioned into hundreds of discrete trap levels, facilitating computationally efficient finite-element analysis while preserving the essential physical features of the original trap profile.

IV. SIMULATION RESULT AND VALIDATION OF THE MODEL

The simulation results of the constructed electron emission model were generated using MATLAB R2024b. By adapting the calibrated Poisson solution in retention state condition, various comparisons and R -square validation over the results were shown in this section. To validate the model over TCAD, some variations of parameters were given and compared. Fig. 5(a) shows the variations in the electron emission time transient curves observed in programmed cells, where the effects of trap concentration variations are incorporated using BETOX trap profiles corresponding to 1k, 10k, and 50k program/erase (P/E) cycles, and the ΔV_{th} values range from 0 to 2 V. Fig. 5(b)–(d) shows the variations of t_{01} from 2 to 4 nm, and Fig. 5(e) and (f) shows the improvement of current model from previous studies [4].

Fig. 5(a) shows the comparison of models across various ΔV_{th} values ranging from 0 to 2 V. It is observed that the more intense the electric field applied to the trapped electrons, the shorter the lifetime for electron emission in the model.

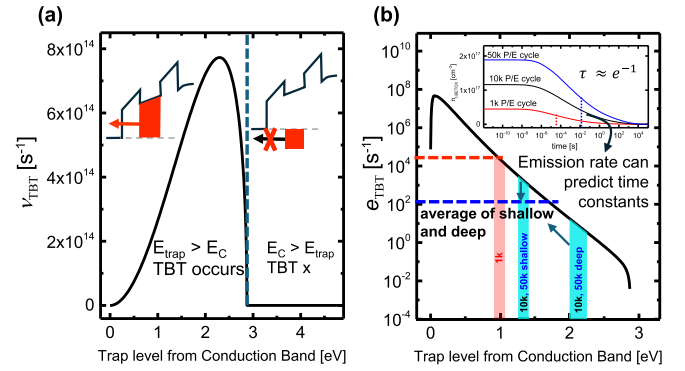


Fig. 6. Visualization of (a) ν_{TBT} computation by (7) and (11) in the case of $\Delta V_{\text{th}} = 1$ V and $t_{01} = 2$ nm and (b) emission rate with time constant prediction.

Although this observation is valid, the difference remains relatively minor compared to other variations.

Fig. 5(b)–(d) shows strong agreement between the proposed model and TCAD results over varying t_{01} from minimum 2-nm values to maximum 4-nm values across different trap profiles indicated in Fig. 4. It shows that the emission behavior is highly dependent on t_{01} about 100-fold difference by 0.5-nm difference.

The proposed model of TBT emission explicitly excludes temperature dependencies. This is due to the nature of the band-trap transition model, which assumes the conservation of the energy level of electron while the transition occurs. The temperature has the potential to make a difference in the results by the occurrence of intrinsic effects such as mobility or Fermi distribution within the Si channel, but cannot affect the band-trap transition model.

The proposed model follows the stretched exponential decaying nature of 3-D NAND flash memories' retention characteristics [8]. Meanwhile, a previous model [4] predicted the time constant of emission by using the attempt-to-escape frequency as a fitting parameter, but cannot simulate the exact behavior of TBT, which shows a low R -square value shown in Fig. 5(e) and (f).

These findings confirm that the proposed TBT emission model accurately simulates electron emission across diverse trap profiles and device scales while achieving simulation times below 1 s. This computational efficiency stems from algorithmic optimization of the computational model, which replaces nonlinear approaches used in TCAD or prior detrapping simulations [4], [9] with a parallel-solving framework for the first-order ODE general solution as (3) and (4).

Fig. 6(a) visualizes the ν_{TBT} for the case of $\Delta V_{\text{th}} = 1$ V and $t_{01} = 2$ nm. The ν_{TBT} increases as the trap energy level becomes deeper, which is consistent with the modeling as described in (7) and (11), with approximately square relation. On the other hand, the transmission coefficient ($|T_C|$) decreases as the trap level deepens by an exponential relation. This leads to an overall reduction in the emission rate due to their multiplicative relationship in (5), which is demonstrated in Fig. 6(b).

Time constant of stretched-exponential decaying nature of emission [8] can be approximated by using the relation of $\tau \approx e^{-1}$. The highlighted regions, labeled as 1k, 10k, and 50k for each shallow and deep, represent the most dominant

trap levels within the trap profile distribution, effectively illustrating this tendency and providing insights into predicting time constants for each cycle's trap profile.

V. CONCLUSION

In this study, an accurate TBT emission model of BETOX was developed. By constructing a band-trap transition framework, the attempt-to-escape frequency (ν_{TBT}) for trapped electrons was derived through physical derivation. The simulation time was minimized by employing the general solution of the first-order ODE, thereby avoiding nonlinear iterations required in conventional approaches. The results demonstrated excellent agreement with TCAD simulations based on experimentally measured trap profiles and exhibited robustness against variations in ΔV_{th} and ϕ_1 thickness within BETOX layer.

APPENDIX

By solving the radial Poisson equation [13] under BETOX 3-D NAND flash memory, which is in cylindrical structure

$$\frac{\partial^2 V(r)}{\partial r^2} + \frac{1}{r} \frac{\partial V(r)}{\partial r} = \begin{cases} -\frac{qn_{t,n1}}{\epsilon_{n1}} & \text{if } r \in n1 \\ -\frac{qn_{t,CTN}}{\epsilon_{CTN}} & \text{if } r \in \text{CTN} \\ 0 & \text{if } r \in \text{else} \end{cases} \quad (12)$$

where the boundary condition is as $V_{o1}(r_{o1}) = V_{\text{ch}}$ and $V_{\text{BOX}}(r_{\text{BOX}}) = V_G$, and the solution is as follows.

Potential Solution:

Radial Field Intensity Solution:

$$\begin{cases} V_{o1}(r) = \frac{K}{\epsilon_{ox}} \ln \frac{r}{r_{Si}} + V_{\text{ch}} \\ V_{n1}(r) = V_{o1}(r_{o1}) + \frac{K_1}{\epsilon_{n1}} \ln \frac{r}{r_{o1}} + \frac{qn_{t,n1}}{4\epsilon_{n1}} (r^2 - r_{o1}^2) \\ V_{o2}(r) = V_{n1}(r_{n1}) + \frac{K_2}{\epsilon_{ox}} \ln \frac{r}{r_{n1}} \\ V_{\text{CTN}}(r) = V_{o2}(r_{o2}) + \frac{K_3}{\epsilon_{CTN}} \ln \frac{r}{r_{o2}} + \frac{qn_{t,CTN}}{4\epsilon_{CTN}} (r^2 - r_{o2}^2) \\ V_{\text{BOX}}(r) = V_{\text{CTN}}(r_{\text{CTN}}) + \frac{K_4}{\epsilon_{ox}} \ln \frac{r}{r_{\text{CTN}}} = V_G - \frac{K_4}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r} \end{cases} \quad \begin{cases} F_{o1}(r) = -\frac{K}{\epsilon_{ox}} \frac{1}{r} \\ F_{n1}(r) = -\frac{K_1}{\epsilon_{n1}} \frac{1}{r} - \frac{qn_{t,n1}}{2\epsilon_{n1}} r \\ F_{o2}(r) = -\frac{K_2}{\epsilon_{ox}} \frac{1}{r} \\ F_{\text{CTN}}(r) = -\frac{K_3}{\epsilon_{CTN}} - \frac{qn_{t,CTN}}{2\epsilon_{CTN}} r \\ F_{\text{BOX}}(r) = -\frac{K_4}{\epsilon_{ox}} \frac{1}{r} \end{cases} \quad (13)$$

where

$$K = \frac{V_G - V_{\text{ch}}}{\alpha} - \frac{qn_{t,n1}(r_{n1}^2 - r_{o1}^2)}{2\alpha} \left\{ -\frac{r_{o1}^2}{r_{n1}^2 - r_{o1}^2} \frac{1}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{2\epsilon_{n1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{o2}}{r_{n1}} + \frac{1}{\epsilon_{CTN}} \ln \frac{r_{\text{CTN}}}{r_{o2}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r_{\text{CTN}}} \right\} - \frac{qn_{t,CTN}(r_{\text{CTN}}^2 - r_{o2}^2)}{2\alpha} \left\{ -\frac{r_{o2}^2}{r_{\text{CTN}}^2 - r_{o2}^2} \frac{1}{\epsilon_{CTN}} \ln \frac{r_{\text{CTN}}}{r_{o2}} + \frac{1}{2\epsilon_{CTN}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r_{\text{CTN}}} \right\} \quad (14)$$

$$K_1 = K - \frac{1}{2} qn_{t,n1} r_{o1}^2, \quad K_2 = K + \frac{1}{2} qn_{t,n1} (r_{n1}^2 - r_{o1}^2) \quad (15)$$

$$K_3 = K + \frac{1}{2} qn_{t,n1} (r_{n1}^2 - r_{o1}^2) - \frac{1}{2} qn_{t,CTN} r_{o2}^2 \quad (16)$$

$$K_4 = K + \frac{1}{2} qn_{t,n1} (r_{n1}^2 - r_{o1}^2) + \frac{1}{2} qn_{t,CTN} (r_{\text{CTN}}^2 - r_{o2}^2) \quad (17)$$

$$\alpha = \frac{1}{\epsilon_{ox}} \ln \frac{r_{o1}}{r_{Si}} + \frac{1}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{o2}}{r_{n1}} + \frac{1}{\epsilon_{CTN}} \ln \frac{r_{\text{CTN}}}{r_{o2}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r_{\text{CTN}}} \quad (18)$$

Based on this solution of the cylindrical Poisson equation, the reciprocal of capacitive constants C_{n1}^{-1} and C_{CTN}^{-1} of (2) is as follows:

$$C_{n1}^{-1} = \frac{(r_{n1}^2 - r_{o1}^2)}{2} \left\{ -\frac{r_{o1}^2}{r_{n1}^2 - r_{o1}^2} \frac{1}{\epsilon_{n1}} \ln \frac{r_{n1}}{r_{o1}} + \frac{1}{2\epsilon_{n1}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{o2}}{r_{n1}} + \frac{1}{\epsilon_{CTN}} \ln \frac{r_{\text{CTN}}}{r_{o2}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r_{\text{CTN}}} \right\} \quad (19)$$

$$C_{\text{CTN}}^{-1} = \frac{(r_{\text{CTN}}^2 - r_{o2}^2)}{2} \left\{ -\frac{r_{o2}^2}{r_{\text{CTN}}^2 - r_{o2}^2} \frac{1}{\epsilon_{CTN}} \ln \frac{r_{\text{CTN}}}{r_{o2}} + \frac{1}{2\epsilon_{CTN}} + \frac{1}{\epsilon_{ox}} \ln \frac{r_{\text{BOX}}}{r_{\text{CTN}}} \right\} \quad (20)$$

By using the radial field intensity solution, the value of F_{eq} for the |TC| value can be found as follows:

$$F_{\text{eq},o1} = F_{o1}(r = r_{Si} + 0.5 \cdot t_{o1}) \quad (21)$$

$$F_{\text{eq},n1} = F_{n1}(r = r_{o1} + 0.5 \cdot t_{n1}) \quad (22)$$

which denotes the assumption of a uniform field within the layer with the field intensity at the centroid of the layer.

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